

North East University Bangladesh

Department of Computer Science & Engineering
(CSE)

Deep Learning Lab
Course Code (CSE-460)

Project Title

**Bangladesh Taka Fake Currency Detection
Using Convolutional Neural Network (CNN)**

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Introduction

Fake currency is a serious problem that affects the economy, businesses, and ordinary people. Counterfeit banknotes can cause financial losses and make it difficult to maintain trust in the financial system. Detecting fake notes manually is often time-consuming and may lead to mistakes.

With the development of Artificial Intelligence (AI) and Deep Learning, it is now possible to build automated systems that can identify fake currency quickly and accurately. Convolutional Neural Network (CNN), a popular deep learning technique, is widely used for image recognition and classification tasks.

This project, "Bangladesh Taka Fake Currency Detection Using Convolutional Neural Network (CNN)," aims to develop a smart system that can detect whether a Bangladeshi currency note is real or fake. The system analyzes images of currency notes and learns important features such as color, texture, watermark, and security thread.

After training with a data set of real and fake currency images, the CNN model can classify a note and provide the prediction result. This project demonstrates how deep learning and computer vision can be used to improve currency verification and help reduce the circulation of counterfeit money.

Problem Statement

In Bangladesh, fake currency notes are sometimes difficult to identify because modern counterfeit notes often look very similar to genuine ones. Traditional methods of detecting fake notes usually rely on manual inspection, which requires experience and can lead to human errors.

Solution

Many people do not have access to specialized currency verification devices, making it challenging to verify the authenticity of banknotes during everyday transactions. As a result, there is a need for an automated, accurate, and cost-effective solution that can quickly distinguish between real and fake currency notes.

Objectives

The main objectives of this project are:

1. To develop a deep learning-based system for detecting fake Bangladeshi currency notes.
2. To use Convolutional Neural Network (CNN) techniques for image classification and feature extraction.
3. To analyze important currency features such as color, texture, watermark, and security thread.
4. To accurately classify currency notes as real or fake based on image input.
5. To reduce the dependency on manual currency verification methods.
6. To provide a fast, reliable, and user-friendly counterfeit currency detection solution.
7. To demonstrate the practical application of artificial intelligence and computer vision in financial security.

Existing Dataset

For this project, a data set of Bangladeshi currency note images is used to train and evaluate the CNN model. The data set contains images of both real and fake currency notes collected from publicly available sources and Google Images (for educational purposes) Kaggle Data set.

Dataset Structure

The data set is organized into two main categories: **Real** and **Fake** currency notes.

```
dataset/
├── real/
│   ├── real_001.jpg
│   ├── real_002.jpg
│   ├── real_003.jpg
│   └── ...
└── fake/
    ├── fake_001.jpg
    ├── fake_002.jpg
    ├── fake_003.jpg
    └── ...
```

Dataset Classes

The dataset contains:

- Training Folder
- Testing Folder

Class	Description
Real	Images of genuine Bangladeshi currency notes
Fake	Images of counterfeit Bangladeshi currency notes

Image Specifications

- Image Format: JPG / PNG
- Image Size: Resized to 128×128 pixels
- Color Mode: RGB
- Number of Classes: 2 (Real and Fake)
- Input Shape: (128, 128, 3)

Technologies Used

Programming Language

- Python

Deep Learning Frameworks

- TensorFlow
- Keras

Machine Learning Models

- Convolutional Neural Network (CNN)
- MobileNetV2
- Support Vector Machine (SVM)
- Random Forest
- Logistic Regression

Frontend Technologies

- HTML
- CSS
- JavaScript

Image Processing Libraries

- NumPy
- Keras ImageDataGenerator

Deep Learning Model

In this project, a **Convolutional Neural Network (CNN)** is used to automatically analyze and classify Bangladesh currency images as **real or fake**. CNN is a deep learning model specifically designed for image processing and visual pattern recognition.

Input Image Processing

Feature Extraction using Convolution Layers

- Edges and borders of the currency note
- Color patterns and textures

- Security threads and watermarks
- Fine details such as micro-text and printing patterns

Pooling Operation

After convolution, pooling layers reduce the size of feature maps while keeping important information. This helps to:

- Reduce computational complexity
- Prevent over fitting
- Focus on the most important features

Fully Connected Layers

After feature extraction, the learned features are passed to fully connected layers. These layers combine all extracted information and learn patterns that distinguish real currency from fake currency.

Output Layer

- Output → **Real Currency**
- Output → **Fake Currency**

Literature Review

Paper No	Paper Title	Models Used	Short Summary
1	Enhanced Counterfeit Detection of Bangladesh Currency through Convolutional Neural Networks: A Deep Learning Approach	CNN (Convolutional Neural Network)	This paper uses CNN-based deep learning model to detect counterfeit Bangladeshi currency by learning visual features like texture, color, and security patterns from images.
2	Bangladeshi Currency Authentication Checking System Using Convolutional Neural Networks	CNN (Convolutional Neural Network)	The system applies CNN to classify Bangladeshi banknotes as real or fake based on image feature extraction and training on labeled currency datasets.
3	BD Currency Detection: A CNN-Based Approach with Mobile App Integration	CNN (Convolutional Neural Network), MobileNet (likely lightweight CNN for mobile deployment)	This paper focuses on real-time fake currency detection using CNN and integrates the model into a mobile application for easy user access.
4	Indian Fake Currency Detection using Image Processing and Machine Learning	Image Processing techniques, SVM, Random Forest, KNN (Machine Learning classifiers)	This paper uses traditional image processing for feature extraction and machine learning algorithms to classify fake and real Indian currency notes.
5	Advanced Methods for Identifying Counterfeit Currency: Using Deep Learning and Machine Learning	CNN, SVM, Random Forest, Logistic Regression, Hybrid ML-DL models	This study compares deep learning (CNN) and machine learning models to identify the most accurate method for counterfeit currency detection.

Conclusion:

Bangladesh's economic development is significantly influenced by banks and other financial organizations. These institutions are in danger from counterfeit money, which might reduce their operating effectiveness and result in financial losses. The economy can be affected by counterfeit money, which might lead to companies or unwary people accepting fake bills. This undermines trust in the currency and may hurt consumer purchasing and corporate dealings. To sum up, the significance of detecting counterfeit cash in Bangladesh is rooted in its effect on the credibility of financial institutions, the monetary system, and the general stability of the economy. Resolving the issue of counterfeit money benefits the nation's financial stability, economic development, and standing abroad. Although it is developed as a prototype, it provides a strong foundation for future improvements such as increasing data set size, improving accuracy, and integrating a real-time mobile or web application.

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